

VG-CLIP: A Visually-Guided CLIP for Zero-Shot Anomaly Detection

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Abstract

Contrastive Language-Image Pre-training (CLIP) has laid a foundation for zero-shot anomaly detection (ZSAD). However, existing methods strive to enhance the robustness of the text encoder to generate semantic guidance, which is misaligned with the visual nature of anomaly detection. To address this issue, we propose a visually-guided CLIP (VG-CLIP) for ZSAD, shifting the optimization focus from the text encoder to visual features. Specifically, we develop a Visual Feature Generation Module (VFGM), which incorporates a Visual Fine-tuning Module (VFM) and a Progressive Fusion Module (PFM). VFM employs a multi-cognitive convolutional structure to enhance visual feature perception and knowledge transfer, while PFM enables progressive fusion of multi-level visual features to fully leverage their collaborative advantages. In addition, we design a Text Correction Module (TCM) that dynamically adjusts the text features using local patch features of test images to refine cross-modal alignment. We evaluate VG-CLIP on 14 industrial and medical datasets. The results show that it significantly improves the perception and localization abilities in visual anomalies, achieving excellent ZSAD performance. The source code is provided in the supplementary materials.

1. Introduction

Contrastive Language-Image Pre-training (CLIP) [25] has revolutionized open-world visual perception tasks with its robust cross-modal alignment and zero-shot generalization capabilities. In recent years, researchers have begun exploring its application to the more challenging task of zero-shot anomaly detection (ZSAD). However, discrepancies in task objectives and domain knowledge make this application exceptionally challenging.

The generation of text prompts and extraction of image features represent core challenges in CLIP-based ZSAD methods. Early studies such as WinCLIP [17] and APRIL-GAN [11] primarily rely on manually constructed prompt templates and basic visual feature extraction strategies,

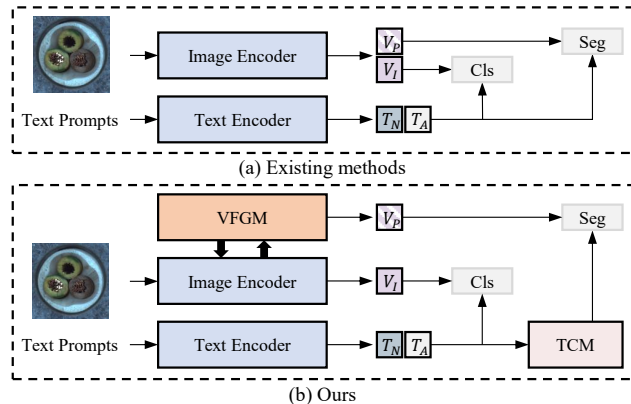


Figure 1. Comparison of existing CLIP-based methods and our proposed VG-CLIP. (a) Existing methods rely on visual-text alignment. (b) Our method aims to generate better visual guidance through the Visual Feature Generation Module (VFGM). Meanwhile, the Text Correction Module (TCM) utilizes this visual guidance to achieve dynamic inference.

thereby establishing the fundamental paradigm for this field. Subsequent works such as AnomalyCLIP [36] introduce prompt fine-tuning into the CLIP text encoder, enabling a shift from manual design to parameterized learning. The latest advancements like Bayes-PFL [24] and AA-CLIP [21] have focused on text-visual joint optimization frameworks, which achieve state-of-the-art (SOTA) performance by collaboratively adjusting dual-modal encoders. This progress reveals a distinct evolutionary trajectory: enhancing ZSAD performance through in-depth text-side optimization. However, in anomaly detection, which is a visually oriented task, this persistent reliance on the text encoder merits reconsideration.

We argue that excessive optimization of the CLIP text encoder is not conducive to the ZSAD task, reflected mainly in two aspects. First, text fine-tuning tends to trigger semantic overfitting, thus undermining cross-domain generalization capabilities [30]. The semantics of text prompts exhibit a strong scenario dependence. For example, “defect” refers to part scratches in industrial scenarios [26], while in med-

058	ical scenarios it corresponds to lesion areas [16]. Text features	111
059	fine-tuned for a specific domain will be overly adapted	112
060	to the semantic expressions of that domain and difficult to	113
061	transfer to other scenarios. Even prompt fine-tuning meth-	114
062	ods that leverage image features [9, 27, 33] can hardly avoid	115
063	affecting generalization capabilities. Second, the ambigu-	116
064	ity of textual semantics can hardly meet the requirement	117
065	of visual precision in anomaly detection. The essence of	118
066	anomaly detection lies in identifying “deviations in visual	119
067	patterns” rather than “differences in semantic categories”.	120
068	Many anomalies such as irregular textures and subtle de-	121
069	formations can be directly discerned through visual features	122
070	but are difficult to describe precisely in text. Even with fine-	123
071	tuning to optimize the text, the semantic information con-	124
072	veyed remains abstract and fails to form an accurate corre-	125
073	spondence with the fine-grained visual features of abnormal	
074	regions, thereby making it difficult for the model to achieve	
075	precise localization of anomalies.	
076	In contrast, optimizing visual features aligns more	
077	closely with the fundamental requirements of ZSAD. First,	
078	abnormal visual patterns exhibit a high degree of common-	
079	ality across different domains. This inherent characteris-	
080	tic allows cross-domain generalization performance to be	
081	enhanced by improving visual representation capabilities.	
082	Second, high-quality visual features can directly encode	
083	fine-grained abnormal details without relying on potentially	
084	misleading textual guidance, thereby providing discrimina-	
085	tive cues for precise localization.	
086	Based on the above analysis, we propose a visually-	
087	guided CLIP for ZSAD, named VG-CLIP. Its core design	
088	philosophy is to adopt an optimization paradigm centered	
089	on visual features, as shown in Fig. 1. To generate high-	
090	quality visual features, we design the Visual Feature Gen-	
091	eration Module (VFGM), which integrates two comple-	
092	mentary sub-components: the Visual Fine-tuning Module	
093	(VFM) and the Progressive Fusion Module (PFM), work-	
094	ing in tandem to refine and integrate visual features at mul-	
095	tiiple levels. Specifically, VFM adopts a multi-cognitive vi-	
096	sual fine-tuning strategy, using multiple convolutional fil-	
097	ters to enhance the feature perception ability of the adapter,	
098	thereby more efficiently transferring visual knowledge from	
099	pre-trained models to anomaly detection tasks. Then, PFM	
100	constructs a multi-level feature fusion architecture to inte-	
101	grate these fine-tuned features from different levels, yield-	
102	ing enhanced features with multi-scale perception capabil-	
103	ities. Additionally, the Text Correction Module (TCM)	
104	adopts a visually guided text correction strategy, dynam-	
105	ically adjusting text features using patch features from a	
106	single test image, enabling precise alignment between text	
107	representations and local image semantics, thereby improv-	
108	ing the accuracy of anomaly segmentation. By integrating	
109	these modules, our VG-CLIP significantly enhances visual	
110	anomaly perception and localization capabilities, achieving	
	a stronger ZSAD performance. Overall, our contributions	111
	are as follows:	112
	• We propose a visually-guided CLIP (VG-CLIP) for	113
	ZSAD, shifting the focus from text encoder fine-tuning	114
	to visual feature enhancement, which aligns with the core	115
	requirement of ZSAD for visual perception.	116
	• We design VFGM that synergizes the multi-cognitive	117
	fine-tuning capability of VFM and the multi-level fusion	118
	power of PFM, aiming to generate high-quality visual fea-	119
	tures. In addition, a simple yet effective TCM is devel-	120
	oped to optimize the segmentation results using features	121
	of the test image itself.	122
	• The comprehensive experiments conducted on 14 real-	123
	world industrial and medical datasets show that our model	124
	exhibits excellent performance in ZSAD task.	125
	2. Related Work	126
	2.1. Classical Anomaly Detection	127
	In the classical anomaly detection paradigm, models are	128
	trained exclusively on normal samples and tasked with iden-	129
	tifying anomalies belonging to the same category during in-	130
	ference. Early statistical modeling methods [28, 34] assume	131
	that normal data follows a specific probability distribution,	132
	identifying anomalies based on the degree of deviation from	133
	this distribution. Similarity-based methods [31, 35] detect	134
	anomalies by calculating the distance or density differences	135
	between data points and their neighboring samples, ex-	136
	celling at capturing local anomalies. With the advancement	137
	of deep learning, representation learning-based approaches,	138
	such as autoencoders [4, 15] and GANs [1, 12], can learn	139
	the latent representations of normal samples, demonstrat-	140
	ing superior performance in handling high-dimensional data	141
	and complex anomalies. However, these classical methods	142
	generally require a large number of normal samples from	143
	known categories for training, failing to address unseen cat-	144
	egories and thus limiting their practical applicability.	145
	2.2. Zero-shot Anomaly Detection	146
	ZSAD identifies anomaly regions without any target class	147
	samples during training. The advent of CLIP has signifi-	148
	cantly propelled the development of this technology.	149
	As a pioneering work, WinCLIP [17] introduces CLIP	150
	into ZSAD, dramatically advancing anomaly segmentation	151
	accuracy through the prior knowledge-enhanced prompt	152
	engineering and multi-scale sliding window technique.	153
	APRIL-GAN [11] achieves precise alignment between vi-	154
	sual patch features and text features via learnable linear pro-	155
	jection layers, effectively overcoming the inefficient win-	156
	dow input limitation in WinCLIP while further enhanc-	157
	ing localization performance. AnomalyCLIP [36] replaces	158
	manual prompt templates with object-independent learn-	159
	able text vectors, and realizes anomaly prediction by cal-	160

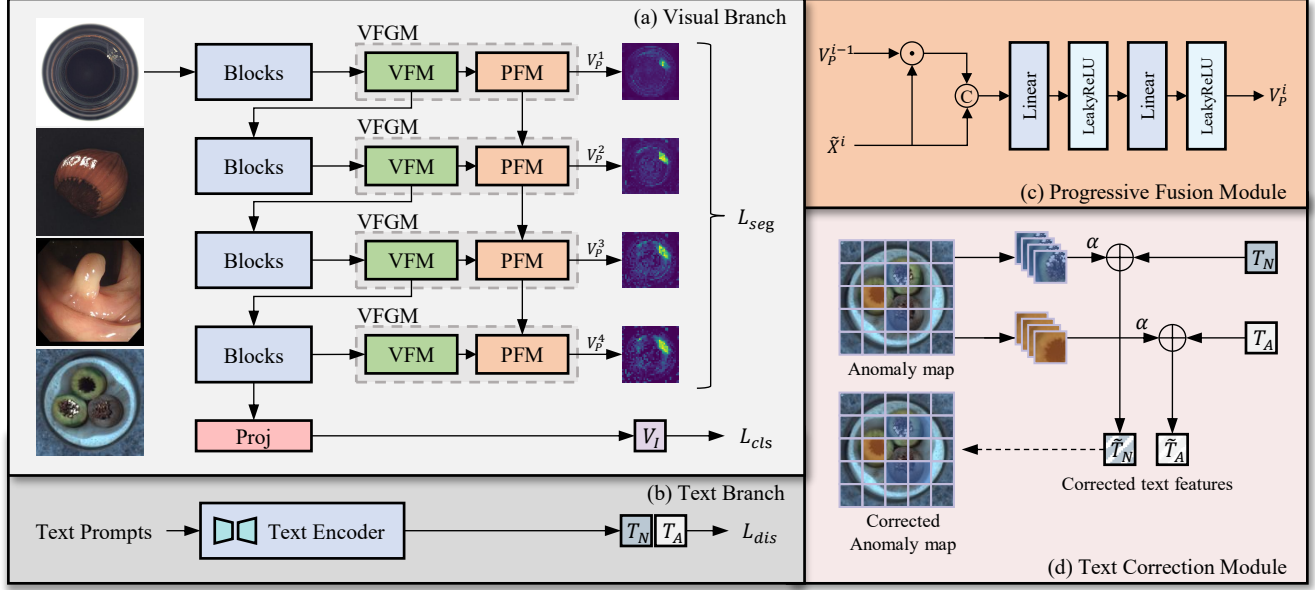


Figure 2. The overall architecture of the proposed VG-CLIP. VG-CLIP employs a dual-branch architecture comprising a visual branch and a text branch. TCM is activated during inference.

culating similarity between multi-level patch features and text features. AdaCLIP [8] improves the alignment of the features by introducing clustering k-means on dense visual features and adding learnable linear projection heads on the CLIP visual encoder. Bayes-PFL [24] models the text prompt space as a learnable probability distribution from a Bayesian perspective, which is used to normalize the text prompt space and effectively improve the model’s generalization for unknown categories. AA-CLIP [21] constructs a clear normal-abnormal semantic boundary in the text feature space through an orthogonal loss function, while enhancing the discriminative ability of CLIP in both visual and text spaces. In contrast to this trend of focusing on text fine-tuning, our method prioritizes optimizing visual features to further enhance ZSAD performance.

In addition, there are some methods based on prior test data. ACR [19] calculates the mean and variance of a batch of test samples and then uses batch normalization to achieve anomaly detection. Based on the assumption that anomaly regions are dissimilar to most regions, MuSc [20] proposes a mutual scoring detection mechanism to improve performance. Despite claiming to be zero-shot methods, these methods require prior access to test data, making them unsuitable for privacy protection or online testing scenarios.

3. Methodology

In this section, we first present the relevant problem definition to help readers better understand the paper. Then we introduce the overall architecture of VG-CLIP. Following this, we describe three core modules of VG-CLIP: VFM,

PFM, and TCM. Finally, we specify the training and inference processes of our approach.

3.1. Problem Definition

Our method adheres to the standard ZSAD task setting, where the model is trained in a supervised manner on an auxiliary dataset and then directly evaluated on datasets from other diverse domains. We define the auxiliary training dataset as $\mathcal{D}_{train} = \{\mathcal{X}_{train}, \mathcal{Y}_{train}, \mathcal{C}_{train}, \mathcal{M}_{train}\}$. Specifically, $\mathcal{X}_{train} = \{x_i\}_{i=1}^M$ denotes training images; $\mathcal{Y}_{train} = \{y_i\}_{i=1}^M$ represents normal or abnormal labels, where a value of 0 for y_i denotes normal and a value of 1 denotes abnormal; $\mathcal{C}_{train} = \{c_i\}_{i=1}^M$ corresponds to category texts; and $\mathcal{M}_{train} = \{m_i\}_{i=1}^M$ denotes ground-truth segmentation masks. Consequently, the test dataset $\mathcal{D}_{test} = \{\mathcal{X}_{test}, \mathcal{Y}_{test}, \mathcal{C}_{test}, \mathcal{M}_{test}\}$ contains data from distinct application domains with unseen anomaly types.

3.2. Overall Architecture

As illustrated in Fig. 2, our proposed VG-CLIP employs a dual-branch architecture comprising a visual branch, a text branch, and a TCM.

In the visual branch, we integrate multiple VFGMs into the intermediate layers of the CLIP visual encoder to fine-tune and fuse visual features. Given an input image $I \in \mathbb{R}^{3 \times H \times W}$, this branch outputs a set of local visual features $\{V_P^i \in \mathbb{R}^{N \times d}\}_{i=1}^4$ for segmentation and a global visual feature $V_I \in \mathbb{R}^d$ for classification. In the text branch, we only insert linear adapters into the first 3 layers of the CLIP text encoder. Given a set of input text prompts, this

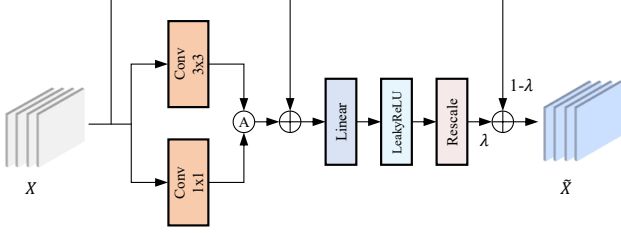


Figure 3. The detailed structure of VFM.

branch produces two disentangled text features, $T_N \in \mathbb{R}^d$ and $T_A \in \mathbb{R}^d$, which are utilized in both the segmentation and classification tasks. More information about the text prompts is provided in the appendix.

Regarding TCM, it is activated exclusively during inference to improve the segmentation performance, and its mechanism will be elaborated on in subsequent sections.

3.3. Visual Fine-tuning Module

Fine-tuning serves as the core technology for adapting visual models to downstream tasks. It enables targeted adjustments to the parameters of pre-trained models, thus facilitating the efficient transfer of general visual knowledge acquired from large-scale datasets. Current mainstream fine-tuning methods for visual models such as Linear Adapter [10] and LoRA [14] largely draw inspiration from adapter designs in Natural Language Processing (NLP). However, visual data differs fundamentally from language data. Directly transplanting NLP adapters thus fails to capture complex visual feature associations, limiting transformation capacity and knowledge transfer efficiency.

To address the limitations of existing methods, we propose VFM. It is tailored to the characteristics of visual data to enhance feature perception and knowledge transfer capabilities. Fig. 3 illustrates the detailed structure of VFM. Specifically, for the input feature X , we first employ a dual-branch convolutional structure for multi-scale feature extraction, capturing diverse visual patterns ranging from local details to global structures:

$$X' = \frac{1}{2}(\text{Conv1}(X) + \text{Conv3}(X)) + X, \quad (1)$$

where Conv1 and Conv3 denote a 1x1 and a 3x3 convolutional layer, respectively. Then, we perform further fusion and nonlinear transformation via a linear layer and an activation function, forming a more universal visual representation. This enhances the model's generalization capability on unseen data:

$$X'' = \text{LeakyReLU}(\text{Linear}(X')). \quad (2)$$

To stabilize the training process and maintain feature consistency, we adjust the scale of X'' via a normalization

operation to match that of the input feature X , formulated as:

$$X''' = X'' \cdot \frac{\text{Norm}(X)}{\text{Norm}(X'')}, \quad (3)$$

where Norm is computed along the feature dimension. Finally, X and X''' are weighted and fused to obtain the enhanced feature $\tilde{X}$, which serves as the input for the next block:

$$\tilde{X} = \lambda \cdot X''' + (1 - \lambda) \cdot X, \quad (4)$$

where λ is a hyperparameter to control the residual ratio, adjusting the fusion degree of knowledge to preserve the original CLIP's generalization ability while improving performance.

3.4. Progressive Fusion Module

In the CLIP visual encoder, features at different levels are complementary. Thus, enabling efficient collaboration between multi-level features is crucial for enhancing model performance. Existing methods [21, 24] typically employ a hierarchical independent mapping strategy, directly outputting features from each intermediate layer of CLIP after linear transformation. These approaches neglect information exchange across levels, making it difficult to fully exploit the collaborative advantages of multi-level features.

In contrast, our PFM promotes full complementarity between shallow details and deep semantics through progressive fusion of adjacent hierarchical features, ultimately generating hierarchical features with stronger representational power. For the local visual feature V_P^{i-1} from the upper level and the output $\tilde{X}$ of the current level VFM, we first apply the Hadamard product and concatenation to obtain the fused feature X' . In this design, the Hadamard product strengthens details and semantic patterns between V_P^{i-1} and $\tilde{X}$ via element-wise interaction, while concatenation integrates these strengthened associative patterns with the original features to provide richer representations for subsequent processing. Formulated as:

$$X' = \text{Concat}((V_P^{i-1} \odot \tilde{X}), \tilde{X}), \quad (5)$$

where $\odot$ denotes the Hadamard product. Then, X' is passed through two linear layers to transform and refine the fused feature representation:

$$X'' = \text{LeakyReLU}(\text{Linear}(X')), \quad (6)$$

$$X''' = \text{LeakyReLU}(\text{Linear}(X'')). \quad (7)$$

Finally, to ensure the stability of feature distribution, we process X''' via a normalization layer to obtain the local visual feature V_P^i of the current level:

$$V_P^i = \text{Normalization}(X'''). \quad (8)$$

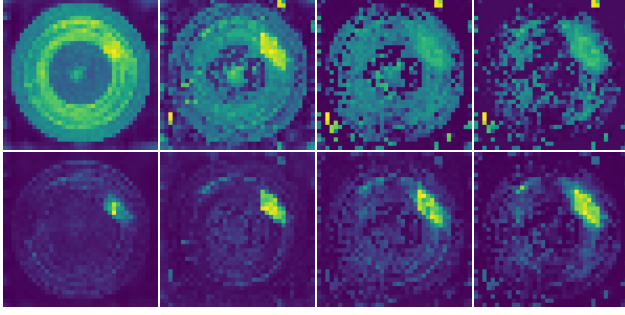


Figure 4. The cosine similarity maps of features across different layers. In these maps, yellow indicates that a patch has lower similarity with other patches, while blue denotes higher similarity. The color distribution can reflect the feature distribution. It is evident that the features enhanced by PFM (lower row) exhibit greater discriminability than initial features (upper row).

To intuitively demonstrate the effect of PFM, we calculate the cosine similarity maps of features at each layer, with the results shown in Fig. 4. It can be observed that, compared to the initial features, the features enhanced by PFM exhibit a more focused and discriminative pattern in terms of similarity distribution. This visual evidence confirms that PFM can effectively unleash the collaborative potential of multi-level features, generating more discriminative feature representations.

3.5. Text Correction Module

The CLIP-based ZSAD methods rely on fixed visual-text alignment mappings and preset text prompt templates during inference, thus adhering to a static feature matching paradigm. This paradigm exhibits notable limitations. On the one hand, text features T_N and T_A fail to adapt to the visual details or contextual variations of specific objects in test samples, leading to biases in feature alignment. On the other hand, when discrepancies exist between the distributions of test and training data, static text prompts struggle to encompass all anomaly patterns, impairing the model’s sensitivity to abnormal regions and its discriminative accuracy.

In response to the above issues, we propose TCM, whose core is leveraging local patch features of a single test image to achieve adaptive correction of text features. Specifically, during inference, we first select high-confidence normal and abnormal patch samples from the current test image. This process relies on the initial anomaly map m_{pred} between the local visual feature V_P and the initial text features T_N and T_A . Formulated as:

$$\mathcal{V}_N = \text{TopkNormalPatchFeatures}(V_P, m_{pred}, k) \quad (9)$$

$$\mathcal{V}_A = \text{TopkAbnormalPatchFeatures}(V_P, m_{pred}, k) \quad (10)$$

where k denotes the Top-k parameter. We then update the initial text features using the average feature representations

Algorithm 1 Text Correction Module (TCM)

Input: Local visual feature V_P , initial text features T_N, T_A , initial anomaly map m_{pred}

Parameter: Top-k parameter k , weight α

Output: Refined text features $\tilde{T}_N, \tilde{T}_A$

```

1: // Initialization.
2:  $\tilde{T}_N = T_N$ 
3:  $\tilde{T}_A = T_A$ 
4: // Check if there are abnormal patches.
5: if  $\min(m_{pred}[:, :, 1] - m_{pred}[:, :, 0]) < 0$  then
6:   // Select top-k normal patch features.
7:    $\mathcal{V}_N = \text{TopKNormalPatchFeatures}(V_P, m_{pred}, k)$ 
8:   // Select top-k or all abnormal patch features.
9:   // The number of abnormal patches may not be sufficient.
10:   $\mathcal{V}_A = \text{TopKAbnormalPatchFeatures}(V_P, m_{pred}, k)$ 
11:  // Update.
12:   $\tilde{T}_N = \alpha \cdot \text{Average}(\mathcal{V}_N) + (1 - \alpha) \cdot T_N$ 
13:   $\tilde{T}_A = \alpha \cdot \text{Average}(\mathcal{V}_A) + (1 - \alpha) \cdot T_A$ 
14: end if
15: return  $\tilde{T}_N, \tilde{T}_A$ 

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of these two types of high-confidence patches:

$$\tilde{T}_N = \alpha \cdot \text{Average}(\mathcal{V}_N) + (1 - \alpha) \cdot T_N \quad (11)$$

$$\tilde{T}_A = \alpha \cdot \text{Average}(\mathcal{V}_A) + (1 - \alpha) \cdot T_A \quad (12)$$

where α regulates the intensity of visual evidence injection. This process enables text features to dynamically anchor the visual feature distribution of the current test image. Based on the updated text features, we recalculate the anomaly map, which significantly improves the accuracy of anomaly segmentation for a single image. And the detailed workflow of TCM is further illustrated in Alg. 1.

Unlike ACR and MuSc, our TCM only uses the information of the test image itself instead of the entire test set, avoiding the problem of data leakage.

3.6. Training and Inference

During the training process, we obtain the anomaly maps $\{m_{pred}^i \in \mathbb{R}^{N \times 2}\}_{i=1}^4$ by calculating the cosine similarity between the local visual feature V_P^i and the text features T_N and T_A :

$$m_{pred}^i = \text{CosSim}(V_P^i, [T_N, T_A]), \quad (13)$$

where $[\cdot, \cdot]$ denotes the concatenate operation. Then these anomaly maps are reshaped and upsampled to $\mathbb{R}^{H \times W \times 2}$ to align with the height H and width W of the ground-truth segmentation mask m . The global anomaly prediction $y_{pred} \in \mathbb{R}^2$ is also calculated in a similar way:

$$y_{pred} = \text{CosSim}(V_I, [T_N, T_A]). \quad (14)$$

Table 1. ZSAD segmentation performance comparison (P-AUROC). The best result is marked in **bold** and the second best in underline.

Model	Dataset	CLIP [25]	WinCLIP [17]	AnomalyCLIP [36]	AdaCLIP [8]	Bayes-PFL [24]	AA-CLIP [21]	Ours
Industrial	VisA	46.6	79.6	95.4	95.1	<u>95.6</u>	95.5	95.9
	MVTec	38.4	85.1	91.1	88.8	<u>91.8</u>	<u>91.9</u>	92.5
	MPDD	62.1	95.2	96.2	96.6	97.1	<u>97.2</u>	97.6
	BTAD	30.6	71.4	93.3	87.7	93.9	97.1	<u>96.7</u>
	KSDD	-	97.9	<u>99.4</u>	96.1	99.6	98.4	97.6
	DAGM	-	83.2	<u>99.1</u>	97.0	99.3	96.7	97.6
	DTD-Synthetic	-	82.5	97.6	94.1	<u>97.8</u>	97.7	98.3
Medical	Brain MRI	68.3	86.0	96.2	93.9	95.5	<u>96.5</u>	96.8
	Liver CT	90.5	96.2	93.9	94.5	96.5	<u>97.9</u>	98.6
	Retina OCT	21.3	80.6	<u>92.6</u>	88.5	95.6	95.6	95.6
	ColonDB	49.5	64.8	81.9	79.3	82.1	<u>83.8</u>	84.2
	ClinicDB	47.5	70.7	85.9	84.3	89.6	90.0	<u>89.7</u>
	Kvasir	44.6	69.8	81.8	79.4	85.4	<u>87.4</u>	87.6
	CVC-300	49.9	67.5	95.4	92.9	<u>96.6</u>	96.5	96.9

Our VG-CLIP is supervised by multiple loss functions, including the segmentation loss $\mathcal{L}_{\text{seg}}$ and classification loss $\mathcal{L}_{\text{cls}}$ for the visual branch, as well as the disentanglement loss $\mathcal{L}_{\text{disentangle}}$ for the text branch. The total loss function is given by:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{cls}} + \mu \cdot \mathcal{L}_{\text{disentangle}}, \quad (15)$$

where μ is a balancing parameter set to 0.1. More details are provided in the appendix.

The difference in the inference process is that we use TCM to optimize the segmentation results. Specifically, we first obtain the corrected text features $\tilde{T}_N$ and $\tilde{T}_A$, and then recalculate the anomaly map to achieve better outcomes:

$$\tilde{m}_{\text{pred}}^i = \text{CosSim}(V_P^i, [\tilde{T}_N, \tilde{T}_A]). \quad (16)$$

The final anomaly score is derived from the weighted sum of multiple $\tilde{m}_{\text{pred}}^i$ after reshaping and upsampling.

4. Experiments

4.1. Datasets

To study anomaly classification and segmentation performance, we conduct experiments on 14 publicly available datasets that cover various industrial inspection scenarios and medical imaging domains (including photography, endoscopy, CT, MRI, and OCT) to evaluate the performance of VG-CLIP. In industrial inspection, we consider VisA [37], MVTec [5], MPDD [18], BTAD [22], KSDD [29], DAGM [13] and DTD-Synthetic [2]. In medical imaging, we consider Brain MRI, Liver CT and Retina OCT from BMAD [3], and four different colon polyp detection datasets with different views (CVC-ClinicDB [7], CVC-ColonDB [6], Kvasir-SEG [23], and CVC-300 [32]). Following previous works, we use VisA as the auxiliary train-

ing dataset. To assess VisA itself, we train our model on MVTec.

4.2. Evaluation metrics and baselines

We use the Area Under the Receiver Operating Characteristic Curve (AUROC) as the metric. We compute the AUROC at both the image and pixel levels to comprehensively assess the effectiveness of the model in detecting and localizing anomalies. The competing methods include CLIP [25], WinCLIP [17], AnomalyCLIP [36], AdaCLIP [8], Bayes-PFL [24], and AA-CLIP [21].

4.3. Implementation details

Following previous works, we use OpenCLIP with the ViT-L/14 architecture as the backbone, and input images are resized to 518×518. All parameters of CLIP remain frozen. For VFGM, we insert it after the [6, 12, 18, 24] layers of the CLIP visual encoder, and set the hyperparameter λ to 0.01. For TCM, we set k to 50 and α to 0.02 for all datasets. We train the model for 30 epochs while setting the learning rate to 5×10^{-4} for the visual encoder and 1×10^{-5} for the text encoder. Parameters are updated by Adam optimizers. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU.

4.4. Quantitative results

Tab. 1 and Tab. 2 present the quantitative results of ZSAD across different datasets in both industrial and medical domains. Our VG-CLIP achieves SOTA performance on both image-level and pixel-level metrics across most datasets, surpassing other comparative methods. Bayes-PFL focuses solely on exploring the distribution of the text prompt space without targeted optimization of visual features, while AA-CLIP merely aligns the visual space with a pre-optimized text space, lacking in-depth mining and adjustment of visual

Table 2. ZSAD classification performance comparison (I-AUROC). The best result is marked in **bold** and the second best in underline.

Model	Dataset	CLIP [25]	WinCLIP [17]	AnomalyCLIP [36]	AdaCLIP [8]	Bayes-PFL [24]	AA-CLIP [21]	Ours
Industrial	VisA	66.4	78.1	82.1	83.0	87.0	<u>84.8</u>	87.0
	MVTec	86.1	91.8	91.5	92.0	<u>92.3</u>	92.0	94.2
	MPDD	73.0	63.6	73.7	72.1	<u>80.2</u>	79.3	80.3
	BTAD	73.6	83.3	89.1	91.6	93.2	94.9	<u>94.6</u>
	KSDD	-	93.5	90.3	95.9	<u>97.3</u>	97.0	98.7
	DAGM	-	89.6	95.6	96.5	<u>97.7</u>	95.8	98.8
	DTD-Synthetic	-	95.0	94.5	92.8	<u>95.1</u>	91.4	96.3
Medical	Brain MRI	58.8	66.5	<u>83.3</u>	80.2	82.2	84.0	82.2
	Liver CT	54.7	64.2	61.6	64.2	61.9	<u>71.4</u>	72.0
	Retina OCT	65.6	42.5	75.7	82.7	83.75	<u>84.0</u>	86.1

Table 3. Ablation analysis of key modules.

Module	Model Variants					Ours
	A	B	C	D	E	
VFM	×	✓	×	×	✓	✓
PFM	×	×	✓	×	✓	✓
TCM	×	×	×	✓	×	✓
P-AUROC	91.3	91.4	91.6	91.5	91.9	92.5
I-AUROC	91.6	93.6	92.8	91.6	94.2	94.2

representations. These limitations result in their relatively weaker performance. In contrast, our method employs targeted visual optimization modules and strategies, which enable further unleashing the discriminative power of visual features, thereby achieving superior generalization performance and detection accuracy. More detailed experimental results are provided in the appendix.

4.5. Qualitative results

Fig. 5 presents the qualitative results between our VG-CLIP and two recent SOTA methods, Bayes-PFL and AA-CLIP, across both industrial and medical datasets. Compared with these methods, our VG-CLIP demonstrates a stronger capability to distinguish between normal and abnormal regions, achieving precise segmentation while reducing false positives in object foregrounds. These results robustly validate the effectiveness of our visual-centric optimization paradigm in capturing abnormal visual patterns.

4.6. Ablation Studies

We conduct ablation studies to demonstrate the effectiveness of our modules by training the model on VisA and testing on MVTec. All other parameters and settings, unless specifically noted, remain consistent with those detailed earlier.

Effects of Individual Modules. Tab. 3 reports the ablation results of seven model variants against the complete

model to quantify the contribution of each module. Specifically, variant A serves as the baseline, which is designed with several linear adapters inserted into the intermediate layers of the CLIP visual encoder. As for other variants, we integrate VFM, PFM, and TCM into our method to dissect their individual roles.

As shown, each module contributes to the superior performance of VG-CLIP. Specifically, VFM enables efficient transfer of visual knowledge from pre-trained models and enhances the model’s perceptual capabilities, thereby boosting classification performance. And PFM functions to fuse multi-level features from the CLIP visual encoder, supplementing fine-grained spatial details and high-level semantic information required for segmentation, which improves both segmentation and classification performance. TCM optimizes segmentation performance by dynamically correcting alignment discrepancies between patch features and text features using visual evidence from the test image itself. Note that the refined text features generated by TCM only affect the segmentation process, so theoretically the classification performance will not improve. More results are provided in the appendix.

Analysis on Text Correction Module. The essence of TCM lies in its dual-path anomaly detection mechanism. The first path generates an initial anomaly map by comparing patch features with text features. The innovative second path establishes a self-contrast detection mechanism: it adaptively learns representative normal and abnormal visual prototypes from the test image, then evaluates each patch against these two internal prototypes to produce another anomaly map. We visualize the abnormal patches selected by TCM in Fig. 6, most of which correspond to the true abnormal patches, demonstrating that the extracted visual prototypes are highly representative and provide effective guidance for anomaly detection.

To evaluate the impact of the Top-k parameter k and the fusion weight α on the segmentation performance in TCM, we conduct a series of experiments with different hyperpa-

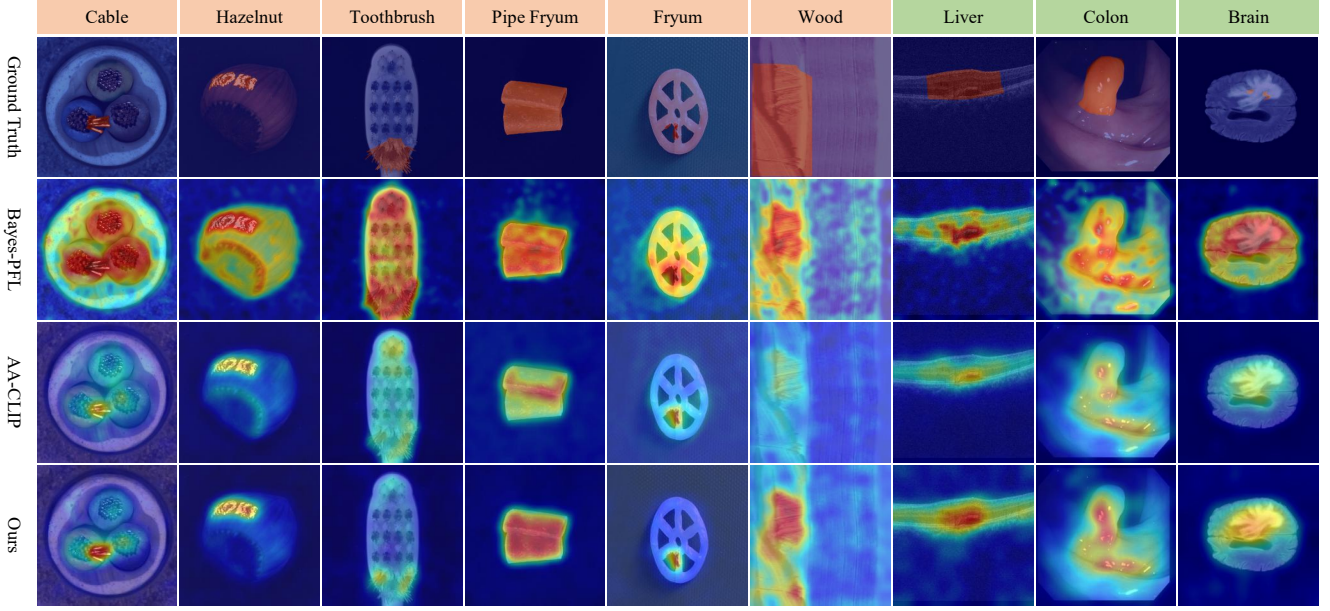


Figure 5. Qualitative results of different methods.

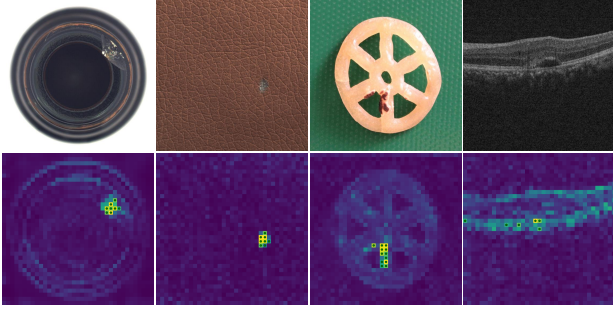


Figure 6. Visual analysis of TCM: Original images (upper row) and their corresponding cosine similarity maps (lower row), with selected anomaly patches marked as green dots.

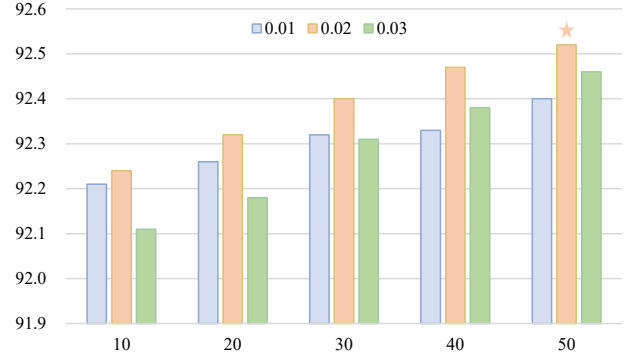


Figure 7. Ablation analysis of hyperparameters in TCM. In this bar chart, the x-axis denotes the parameter k , the y-axis represents the evaluation metric P-AUROC, and bars of different colors correspond to the parameter α .

parameter combinations, as shown in Fig. 7. Here, k controls the number of selected high-confidence patch features: increasing k incorporates richer visual information to complement the text features, but an excessively large k tends to introduce redundancy and noise, degrading performance. Meanwhile, α regulates the degree to which visual evidence adjusts the text guidance. A value that is too low weakens the corrective effect from patch features, while one that is too high amplifies interference. The optimal combination, where k is 50 and α is 0.2, achieves an effective integration of visual cues into the text features, thereby enhancing the overall segmentation accuracy. It should be noted that the optimal parameters are obtained on MVTec, and adjusting them based on specific datasets may yield better performance.

5. Conclusion

In this paper, we propose VG-CLIP, a novel visually-guided framework for ZSAD that challenges the prevailing text-centric optimization paradigm. We first analyze the limitations of excessive text encoder optimization, highlighting the superiority of visual-centric approaches for capturing abnormal patterns. Then we design a comprehensive visual enhancement framework comprising VFGM for generating high-quality visual features through multi-cognitive fine-tuning and progressive fusion, along with TCM for dynamically correcting text features using visual evidence. Extensive experiments on diverse industrial and medical datasets demonstrates that our approach achieves excellent ZSAD performance.

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